Computer Architecture and C

A look beneath

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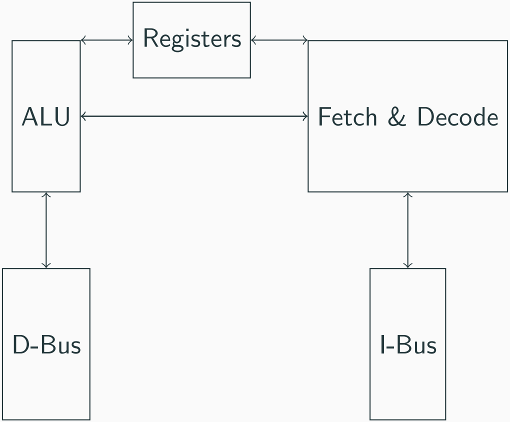
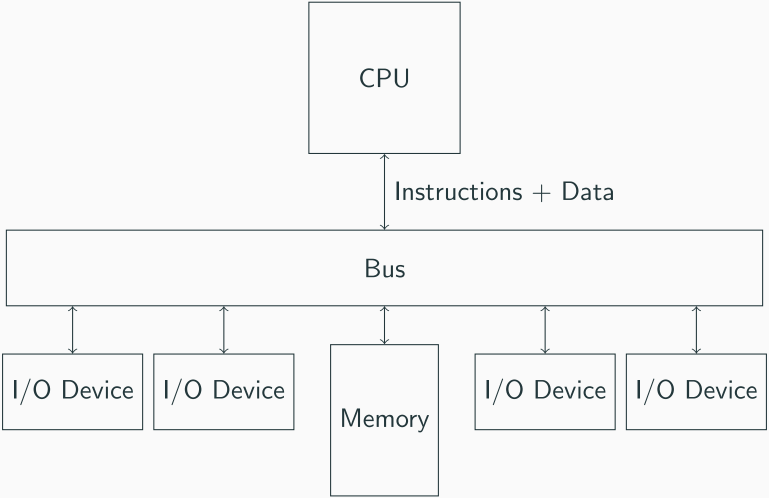
Intro

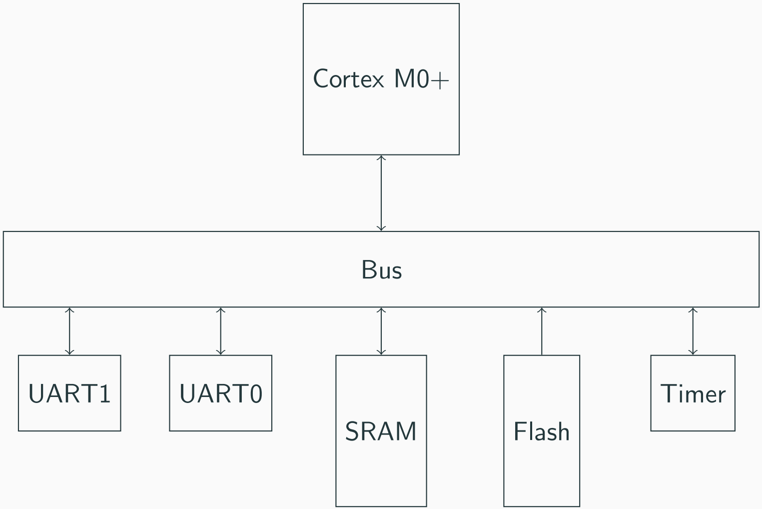
CPU Structure



Von Neumann Architektur



Example implementation



Execute Instruction

- Fetch at Program Counter using I-Bus
- Decode Instruction
- Access Operands
- Execute

Example

mov r1,r2

Instruction fetches

Instruction fetches may stall the CPU.

Load Instruction

- Fetch at Program Counter using I-Bus
- Decode Instruction
- Access Operands
- Read data from RAM

Example

ldr r1, [r2]

Memory Writes

Memory Reads may stall the CPU

Store Instruction

- Fetch at Program Counter using I-Bus
- Decode Instruction
- Access Operands
- Write Data to Store Buffer

Example

str r1, [r2]

Memory Writes

Memory Writes may not be executed immediately.

Reason?

- RAM/ROM can be slower than fcpu
- Access may needs longer Time
- Different Bus master Accesses same Bus

Bus

Master Interface

- Read Request
- Write Request
- Address Based
- Word Size

Slave Interfaces

- Read Response
- Write Response
- Mapped on Address Range
- Word Size

Bus as Multiplexer

- Maps Requests for Slave Address Range to Slave
- Multiple Channels possible (Multi Master)

STM32g070 Reference Manual